Highlights

- Beta-band waveform asymmetry declines with age (Cohen's d = 0.7), replicated across two independent EEG cohorts (N > 800)
- Slow and fast oscillations show opposing asymmetry with a spatial double dissociation between posterior alpha and central beta
- Spatial organization of waveform asymmetry simplifies with age: magnitude increases while diversity decreases
- Theta spatial entropy predicts memory and executive performance independent of age
- Waveform asymmetry provides a rapid biomarker accessible from standard 3-minute EEG recordings